

Alejandro Sánchez Yalí

AI Engineer – Agents & Automation



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SUMMARY

AI Engineer with 8+ years of back-end engineering in Python and 3+ years building LLM agents and automations end to end. Builds agentic workflows with LangGraph and MCP, async back-end services with FastAPI and Celery, and the integrations that give agents the context they need. Uses Claude Code daily as the primary development workflow and has built internal tooling that feeds coding agents richer project context. Takes automations from proof of concept to production: an AI content-generation pipeline built at Lapzo is in use with real users. M.Sc. in Mathematics; based in Antioquia, Colombia.

TECHNICAL SKILLS

- **Agents & LLMs:** LangGraph, LangChain, MCP, LiteLLM, RAG, OpenAI API, Whisper, ElevenLabs
- **AI-assisted engineering & automation:** Claude Code (1.5+ years, daily), n8n, Celery background workers
- **ML & Mathematics:** PyTorch, TensorFlow, Scikit-learn, JAX, Flax, NumPy, Pandas, linear algebra, probability, statistical inference
- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Backend:** FastAPI, Django, Celery, Flask, NestJS, Node.js, RESTful APIs, microservices, event-driven design
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Architecture:** Clean architecture, DDD, SOLID, modular monolith, screaming architecture

EXPERIENCE

AI Specialist

Lapzo

Aug 2025 – Sep 2026

Remote, Mexico

- Built an AI-powered educational content generator, now in use with real users, with LangChain, LangGraph and n8n, with Celery workers running the generation pipelines in the background
- Built Lapzo-Tools, internal tooling for reviewing implemented code and giving coding agents richer project context
- Built an HR agent (LangGraph, FastAPI, PostgreSQL) that answers company-policy questions with citations and executes HR requests with human confirmation, with per-tenant isolation; shipped to QA and staging with a runbook and a handover
- Moved the embeddings pipeline from n8n into the platform and migrated the course-vectorization system to a dedicated VM; automated pre-invoicing workflows with n8n
- Closed 57 tasks and 6 epics in the first nine months of 2026 with a single production bug (per the team's ClickUp board)

Independent Software & AI Consultant

Freelance

Aug 2024 – Present

Remote

- Use Claude Code as the primary day-to-day development workflow on live client projects, steering the agent to break down problems and critically reviewing its output to ship end-to-end features without eroding architectural boundaries
- Built a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP integrations with automatic human escalation; lead engineer of Aluna (see below)
- Led technical design sessions with clients, translating business requirements into system architectures and naming the trade-off behind each technology decision
- Designed database architecture for a client's workflow automation platform and served as code reviewer for their development team

Full Stack Engineer

Monadical

Apr 2021 – Aug 2024

Remote, Canada/US

- Built and integrated AI services for information retrieval with large language models (OpenAI, LangChain, Whisper) in production web applications
- Participated in system design and architectural decisions, evaluating trade-offs across performance, scalability and maintainability

- Built and scaled microservices for AI/ML and data-intensive applications; designed ETL pipelines and custom Django ORM extensions
- Did code reviews on a distributed team across Canada, the US and Latin America

Full Stack Engineer
BCFort

Aug 2018 – Oct 2020
Medellín, Colombia

- Designed system architecture for blockchain platforms: smart contract patterns, data models and service boundaries
- Developed blockchain and analytics platforms with Python (Django), Node.js, React and Web3.js on AWS, GCP and IBM Cloud

Professor of Mathematics & Machine Learning
Universidad de Antioquia

Jan 2010 – Oct 2021
Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python, NumPy, Pandas, TensorFlow, PyTorch and Scikit-learn
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research in ML and data analysis

FEATURED PROJECTS

- [Plixiq – AI Customer Support on WhatsApp](#)  – *Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket*
 - Sole architect and engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
 - Modular monolith with 14 bounded contexts enforced in CI; multi-provider LLM layer (Groq default, OpenAI fallback) so switching providers is a config change
- [Aluna – AI Recruitment Platform for Staffing Agencies](#)  – *Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API*
 - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
 - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email
- [Digital Human – Interactive AI Avatar](#)  – *React, Three.js, React Three Fiber, OpenAI GPT, Whisper, ElevenLabs*
 - Open-source conversational avatar with 464 stars and 110 forks: real-time speech-to-text, generation, voice synthesis and lip synchronisation
 - The engineering problem was latency budgeting across four services, since a human notices any pause in a conversation

OPEN SOURCE

- [talking-avatar-with-ai](#)  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- [Monadical-SAS/Morpheus](#)  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- 493 GitHub stars across own open-source projects – top 4.8% of ranked GitHub profiles (GitRanks)

CONFERENCE TALKS

- [Scaling Learning Models: Parallelism Strategies in JAX with Flax](#)  – PyCon Colombia, 2024

EDUCATION

M.Sc. in Mathematics
Universidad de Antioquia

Aug 2010 - Sep 2013
Medellín, Colombia

B.Sc. in Teaching Mathematics and Physics
Universidad de Antioquia

Jan 2004 - Oct 2009
Medellín, Colombia